YUSUF ABOZEID

AI & Embedded Systems Engineer | NLP, LLMs & Edge AI

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SUMMARY

AI and embedded systems engineering student building NLP, LLM and Edge AI systems. 1st place at the National AI Hackathon (EVA-AI, an evidence-grounded clinical RAG platform) and Top 6 national finalist at RoboDam 2026 (RoboApex, an ESP32 edge-AI microgrid gateway). Experienced in fine-tuning transformers (DeBERTa-v3, AraBERT), hybrid retrieval, and production REST APIs.

EDUCATION

B.Sc. in Computer Science, Major in Artificial Intelligence — Future University in Egypt (FUE) 2023 – Expected 2027

GPA: 3.38 / 4.00 | Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Embedded Systems, Data Structures & Algorithms, Database Systems

EXPERIENCE & TRAINING

Microsoft Machine Learning Track — Digital Egypt Pioneers Initiative (DEPI) In Progress

Freelance AI/Python Developer — Kafiil Platform | *Digital Egypt Youth Program* Oct 2025 – Feb 2026

- Built a collaborative-filtering product recommendation engine for an e-commerce platform.
- Implemented an OpenCV/Python vehicle detection tool, delivered within 24 hours.
- Built an Arabic customer-review sentiment classifier producing label distributions and analytics.

PROJECTS

RoboApex — Autonomous Microgrid Gateway & Edge AI | *Lead Embedded & AI Systems Engineer* Aug – Sep 2026

- Built a decentralized microgrid gateway and home energy management system on distributed ESP32 nodes via ESP-NOW peer-to-peer messaging: sub-3ms latency, offline-resilient, no Wi-Fi router needed.
- Implemented an on-device Edge AI “neural cortex”: predictive thermal modeling (dT/dt) forecasting room temperature 30 min ahead, PMV comfort scoring, and real-time current anomaly detection.
- Designed dynamic peak shaving that sheds high-power EV charging loads during grid stress (150-unit hysteresis deadband) to protect solar inverters and household infrastructure.
- Developed an embedded SCADA dashboard with Groq LLaMA 3.3 70B for natural-language telemetry queries and device actions.

EVA-AI — Evidence-Grounded Clinical Decision Support | *Lead AI Engineer & Architect* Aug 2026

- Architected a clinical decision-support RAG system over official NICE NG243 guidelines, achieving ≥80% Hit-Rate and ≥0.82 Precision@K on an 18-query benchmark.
- Built a hybrid retrieval pipeline fusing dense ChromaDB vectors and Okapi BM25 via Reciprocal Rank Fusion (k=60), followed by cross-encoder reranking (ms-marco-MiniLM-L-6-v2).
- Designed a 5-layer safety workflow: sub-3ms pre-retrieval drug/dosage refusal, domain-boundary enforcement, and a zero-network local embedding fallback (BAAI/bge-small-en-v1.5).
- Shipped the full-stack app: FastAPI 0.115, Next.js 15 (TypeScript) with English/Arabic RTL parity, and Sentry tracing with automated PHI/PII sanitization.

Fact Forge — Fake News & Misinformation Detection | *Core Model Developer* May – Jun 2026

- Built an end-to-end fake news detection pipeline reaching 95.6% accuracy on the WELFake dataset (72,000 news records).
- Fine-tuned DeBERTa-v3 with custom weight decay and dropout to curb overfitting and improve generalization.
- Exposed the model through Flask REST endpoints consumed by a React (Vite) dashboard for real-time credibility scoring.

Mesdaq AI — Arabic Misinformation Classification | *Core Model Developer* Jan 2026

- Fine-tuned AraBERT to classify Arabic news articles and mitigate misinformation.
- Designed an Arabic NLP preprocessing pipeline handling morphological complexity, diacritics, and normalization.

HONORS & AWARDS

Top 6 National Finalist — RoboDam 2026 National Hackathon | Damanhour University & ITI Sep 2026

- Top 6 of 65 university teams; demoed RoboApex live before the Minister of Higher Education & Scientific Research.

1st Place — National AI Hackathon | Orange Digital Center, CREATIVA, ITIDA, TIEC & INSTANT Aug 2026

- Won 1st place among top competing teams with EVA-AI, an evidence-grounded clinical decision-support platform.

TECHNICAL SKILLS

Programming: Python, C/C++ (ESP32), SQL, TypeScript, Java, HTML/CSS | **Spoken:** Arabic (Native), English (Professional)
GenAI & NLP: RAG, LLMs (Claude Sonnet 4.6, Gemini, LLaMA 3.3 70B), prompt engineering, citation-grounded generation, Hugging Face Transformers, fine-tuning (DeBERTa-v3, AraBERT), dropout/weight decay, Arabic NLP, sentiment analysis
Retrieval: hybrid search (BM25 + dense), RRF, cross-encoder reranking, ChromaDB, multi-tier caching, Precision@K, Hit-Rate
ML & Vision: PyTorch, OpenCV, object detection, CNNs, collaborative filtering
IoT & Edge: ESP-NOW mesh, SCADA web server, Edge AI, dynamic peak shaving, ADC sensor interfacing
Frameworks & DevOps: FastAPI, Flask, Next.js 15, React (Vite), Sentry, Git, GitHub Actions (CI/CD), Docker, Linux

CERTIFICATIONS & LEADERSHIP

- NVIDIA DLI Certificate — Generative AI & Deep Learning | ITI
- Responsible & Trustworthy AI — SAS Institute
- AI for Business Specialist — National Telecommunication Institute (NTI), passed with 96%

Co-organizer, GDG FUE Chapter — ran AI/ML workshops and hackathons for 200+ students 2024 – 2025